

Physics Cheat Sheet

Topic / Course Name

Mechanics

Newton's second law:

$$\vec{F} = m\vec{a}$$

Kinematic equations:

$$v = v_0 + at$$

$$x = x_0 + v_0t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a\Delta x$$

Energy

$$K = \frac{1}{2}mv^2$$

$$U = mgh$$

$$W = \int \vec{F} \cdot d\vec{r}$$

Electromagnetism

Maxwell's equations (SI):

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Quantum Mechanics

Schrödinger equation:

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi$$

Bra-ket notation:

$$\langle \psi | \psi \rangle = 1$$

$$\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle$$